roblem Statement (প্রায় 2 মিনিট)

Slide শুরু করার লাইন:

“Now I will discuss the core problem our research addresses.”

🔹 Data Dependency

বলবে:

- Most high-performance change detection models rely on multispectral or very high-resolution satellite data
- এসব dataset expensive এবং publicly accessible না
- Developing regions এর জন্য এটি impractical

👉 Key line:

“So the first issue is accessibility of data.”

🔹 False Positives

বলবে:

- Lighting variation
- Atmospheric noise
- Seasonal vegetation change

Traditional models এগুলোকে structural change হিসেবে ভুল detect করে।

👉 বলবে:

“This increases false positives and reduces reliability.”

🔹 Geographic Overfitting

- Models trained on specific city

- New unseen city তে performance drop
- Architectural diversity handle করতে পারে না

👉 বলবে:

“This limits generalization capability.”

🔹 The Gap

শেষে strong conclude করবে:

“Therefore, there is a critical need for a cost-efficient, RGB-based spatiotemporal change detection framework that is robust and generalizable.”

🟢 Objective (Easy Speaking Version)

Start:

“Based on the challenges discussed earlier, we defined the following objectives.”

◇ Framework Development

“First, we aim to develop a deep learning-based framework for building change detection using multi-temporal RGB images.”

“Our focus is to detect building changes accurately using only standard RGB data.”

◇ Benchmark Training

“Second, we train the model on multiple urban benchmark datasets.”

“This helps improve learning across different city patterns and improves cross-region generalization.”

◇ Enhance Robustness

“Third, we aim to improve robustness.”

“The model should handle lighting noise, seasonal variations, and reduce false positive detections.”

◇ Practical Deployment

“Finally, we focus on practical deployment.”

“The system uses only RGB imagery, making it cost-effective and suitable for real-world applications, especially in resource-limited regions.”

Ending Line (Confident Tone)

“These objectives directly address the limitations identified in existing approaches.”

● Literature Review (Easy & Clear Version)

Start:

“Before designing our framework, we carefully reviewed existing research in building segmentation and change detection.”

Minaee et al. (2022)

- This was a survey paper.
- They compared major segmentation models like FCN, U-Net, DeepLab, and Transformer-based approaches.

বলবে:

“This study helped us understand the strengths and limitations of different segmentation architectures and guided our model selection.”

Yu et al. (2022)

- They proposed an attention-based U-Net.
- The attention mechanism helps the model focus more on important building regions.

বলবে:

“This showed that adding attention can improve feature selection in complex urban scenes.”

Liu et al. (2024)

- They proposed a hybrid CNN and Transformer model.
- CNN extracts local features, while Transformer captures global context.
- It performed well in complex urban environments.

বলবে:

“This work demonstrated the benefit of combining local and global feature learning.”

Liu et al. (2025)

- Focused on recognizing building groups instead of individual buildings.
- Used spatial relationships and semantic similarity between buildings.

বলবে:

“This approach improved structural understanding at a higher level.”

🔴 Critical Insight (Very Important Part)

এখানে একটু pause দিয়ে confidently বলবে:

“However, most of these works focus mainly on improving model architecture. They do not specifically address cost-efficient RGB-based deployment or strong cross-city generalization.”

তারপর immediately বলবে:

“This gap motivated us to design a robust and practical RGB-based framework.”

Methodology (Easy Version)

Start:

“Now I will explain our proposed methodology.”

Architecture (Simple Explanation)

- We used a U-Net model.
- It has two main parts: encoder and decoder.

Explain like this:

- The encoder extracts important features from the image.
- The decoder rebuilds the image to generate the final segmentation map.
- Skip connections help keep fine details like building edges clear.

Then say:

“This structure helps the model understand both the overall structure and the exact boundaries of buildings.”

Loss Function (Simple Explanation)

- We used two loss functions: Binary Cross Entropy and Dice Loss.

Explain simply:

- Binary Cross Entropy helps classify each pixel as building or background.
- Dice Loss helps when building pixels are much fewer than background pixels.

Then conclude:

“Combining both losses makes the training more stable and improves segmentation accuracy.”

🟡 Data Collection (Easy Version)

Start:

“To make the model strong and generalizable, we used multiple benchmark datasets.”

Inria Dataset

- High-resolution RGB images
- Urban areas from the US and Austria

WHU Dataset

- Large-scale dataset
- Very accurate building masks

Massachusetts Dataset

- Used mainly for testing
- Helps check if the model works well on new cities

Preprocessing (Simple)

- All images were resized to 512 by 512
- Pixel values were scaled between 0 and 1

Then say:

“This helps the model train smoothly and converge faster.”